

Detection of Exoplanet Systems In Kepler Light Curves Using Adaptive Neuro-Fuzzy System

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Abstract—The paper investigates the usability of fuzzy inference systems, namely ANFIS to assist in the detection of exoplanet transit events from time-series data of Flux intensity values from host stars. We use statistical properties of time curves, principal component analysis, and dynamic time warping (DTW) to extract features from time-series data. We propose the use of aggregated DTW distance to all positive training samples as a feature and show that it significantly increases performance if used along with statistical features and principal components. We propose that fuzzy methods can be used to classify exoplanet systems with a relatively low false positive rate and thus, can be used to filter out possible flux signatures for review by human experts.

Keywords—ANFIS, Time Series, Classification, Kepler, Dynamic Time Warping, Principal Component Analysis, exoplanet.

I. INTRODUCTION

To determine whether life exists anywhere in the universe (other than our home planet) has enormous scientific and philosophical implications. To answer that question, astronomers, astrophysicists and astrobiologists have designed and conducted many experiments. A large subset of these experiments involve recording and analyzing electromagnetic radiation data of some kind and developing techniques to infer relevant information from that data. The Kepler Space Telescope is one of the more well-known of these Experiments.

During its primary mission from 2009-2013, Kepler has recorded photometric observations of more than 200,000 stars [1] and has provided us with insights about the frequency of small planets in the inner solar systems. Finding whether or not planets exist around a star and their distance from their parent star is crucial in determining the possibility of life in those planets. More specifically, a planet needs to be in a distance range known as the Goldilocks zone to be suitable for sustaining life [2].

However, it is constantly recording new data in larger volumes, which is only possible to analyze semi or fully automatically with the assistance of algorithms. Some works have been done to use machine learning to automatically filter out light curves without transition signatures from the Kepler data using different approaches. Random Forests were used in The Autovetter Project [3], while Thompson et al.[4] and Armstrong et al.[5] used unsupervised clustering to group similarly shaped curves. Despite the ever-increasing amount of work on Kepler data, Fuzzy Inference and classification

techniques have not been explored much in the automated vetting of light curves

In this paper, we present a pipeline for automatically detecting light curves that contain signatures of transiting exoplanets. Our pipeline extracts features from the Kepler time Series brightness curves of individual stars. Then, we use the extracted features as inputs to an Adaptive Neuro-Fuzzy Model (ANFIS) that is trained on a set of light curves labeled by human experts.

We test the accuracy of the ANFIS model for different representations of continuous data. Time-series data are found in almost every domain and finding a feature set to better represent a time-series curve is of special interest. We compare the use of statistical features, principal components and a combination of them along with aggregated DTW distance and show that using DTW as a feature improves the accuracy of our ANFIS model by 7%. This finding can be beneficial for use of ANFIS on higher dimensional datasets.

We investigate our neural-fuzzy model's ability to infer the minor changes in brightness signatures that exist between actual transit events and false positives due to instrumental artifacts, stellar variability and other natural or instrumental phenomena and show that although it has relatively low accuracy, it's low false positive rate can be an advantage for its use to flag possible transition events.

II. RELATED WORKS

The research work by Jin and Glass [6] is based on the classification of stellar images measured in light curves over time. To acquire the data, NASA has their own Kepler Spacecraft that took images of thousands of stars across the universe. Certain necessary preprocessing and feature extraction steps were taken before the passing data into different classifiers (Naïve Bayes, Logistic Regression and SVM). The highest classification accuracy of the light curve time series data obtained was 93% using the Logistic Regression method.

In [7], Ryabko and Mary introduced a binary classification technique and the means of clustering on the time-series data. The dataset used for this research was from the BCI competition III. Two other general classification models: Kernel Principal Component Analysis (KPCA) and Support Vector Machine (SVM) were also used to classify the same data for comparison. The accuracy obtained from the proposed clustering method showed relatively better results on 2 out of 3 subjects provided in the dataset.

In another related work [8], Rakthanmanon et al represent a fast approach to classify long time-series data. They initially discussed the efficient role of Dynamic Time Warping (DTW) and how it was used to find similarities between two time-series data. However, when DTW was coupled with the nearest neighbor classification (NN), the adverse effect of it was stressed on its time complexity. For a different approach, the researchers of this proposed work applied Matrix Factorization (MF) to reduce the latent dimensionality of the time-series data. Then they passed the data to Support Vector Machine (SVM) for classification. Finally, it was experimentally proven that MF and SVM took less classification time than just SVM, DTW and DTW and NN together.

Nanopoulos et al. [9] demonstrated the use of statistical feature extraction techniques for classifying time-series data. The authors of the research paper had shown a comparison of classification on Multilayer Perceptron (MLP) Neural Network (NN) upon applying statistical feature extraction methods and otherwise. It was certainly concluded after several experimental observations that statistical feature extraction provided better classification results than without it.

Dynamic Time Warping determines the similarity between time series data and it has been exploited by researchers with much success[8]. Although it is more prominent in the speech recognition domain, it has been successfully used by experts in a wide range of fields -from robotics to medicine, biometrics, music/speech processing to climatology and from aviation, gesture recognition to user interfaces and industrial processing [8].

A research paper on the application of DTW for discovering patterns in real time shows that searching time series data having a length of 421,322 takes more than 3 hours on a single core system and around 25 minutes for an 8-coresystem[10]. However, using their ideas, Rakthanmanon et al. [8] showed that a dataset of similar size can be searched on a single core within one second.

The encouraging performance of DTW while using simple nearest neighbor algorithms to classify time-series data naturally leads to experimentation with more complex machine learning algorithms for performance improvements [11]. However, to apply DTW in such cases, a time-series needs to be represented as a collection of features or a kernel needs to be defined between the curves being compared. The fact that DTW is a measure of difference between two time-series prohibits its direct usage as a feature [11].

It is possible to use DTW to extract features from time curves. In [11] the authors calculated distance through DTW and used as features on classification tasks. They experimentally showed that on 31 out of 47 UCW benchmark datasets, DTW features had improve performance [11].

III. BACKGROUND

A. Fuzzy Logic

Fuzzy logic is a concept of many-valued logic, where unlike Boolean logic (where only binary truth values are allowed), a statement or variable can have a truth value ranging from 0 to 1 [12]. Although fuzzy logic is present in academia since early 20th century, the term 'fuzzy logic' was introduced in 1965 by

Lotfi Zadeh[13-14]. Fuzzy logic has applications in fields ranging from artificial intelligence to control theory.

B. Artificial Neural Networks

An Artificial Neural Network (ANN) is a computation system that has a structure somewhat resembling a biological brain. ANNs have units of computation called neurons that are connected to each other in a systematic way much like their biological counterpart.

Although ANNs have been studied and applied for decades, the recent flourish of computing power and available data has seen a manifold increase in their use in academia and industry.

Neural Networks are comprised of layers of neurons that serve different computational functions. For example, in a classification task, the first layer of an ANN receives the input data and the last layer produces labels. It is the task of the in-between layers, also known as hidden layers to generate representations of the input data. Neural networks are powerful computational tools that are often described as universal function approximators [15].

C. Adaptive Neuro Fuzzy Inference System

Adaptive Neuro Fuzzy Inference Systems or ANFIS in short was proposed by Jang and Sun [5] to deal with parameter identification. Essentially, ANFIS is a realization of sugeno-type fuzzy inference systems that have learning capabilities like ANNs[16].

Fuzzy systems are known to be more interpretable because of their human like reasoning through fuzzy if-then rules. But they lack the learning capability of neural networks. By combining the two, neuro-fuzzy models learn from data and produce interpretable if-then rules which is more preferable in many use cases and domains like medical diagnosis.

D. Dynamic Time Warp

Dynamic Time Warp or DTW is a measure of similarity between two time-series sequences. Unlike conventional distance measures, DTW is robust to phase changes and even length of time-series.

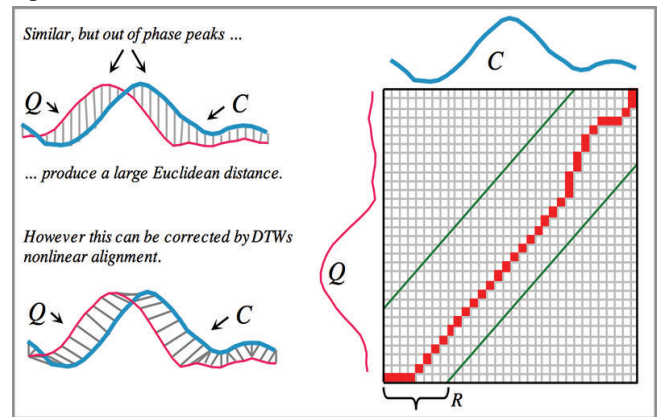


Fig. 1. Advantage of Dynamic Time Warp (from[8])

For example, two sine curves that are out of phase may produce very high distance measures in conventional techniques similar to Euclidean distance (Fig. 1).

On the other hand, Dynamic Time Warping 'aligns' two curves before measuring their distance. Euclidean distance can

be considered as a specific case of DTW where two curves in consideration are perfectly in phase. This can be visualized with the following graph from Rakthanmanon et al. [8].

IV. DATA DESCRIPTION

The obtained data provides details about the variation in flux (light intensity) of several thousand stars, with each star having a binary label of 2 or 1. Here, stars that have an orbiting exoplanet around them are labeled as '2' as confirmed by human experts . The negative examples are labeled as '1'.

The planets do not emit any light, so the light source comes from the stars that orbit around them. When the mentioned star is observed over a long stretch of time- such as many months or years, there may be a pattern-wise ‘dimming’ of the flux (light intensity) at regular intervals. This is proof that there is a planetary body around which the star orbits, so the star can be thought of as the ‘candidate system’, for example, based on the

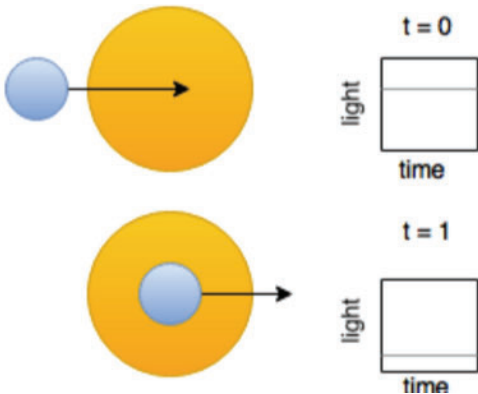


Fig. 2. Drop in flux intensity during transition [17].

different wavelengths of light that are picked up by a satellite, it can lay foundation for ‘confirming’ the candidate system.

Fig. 2 illustrates how light intensity measurements indicate transition events. In the Fig. 2, light (in flux unit) intensities are compared between two scenarios- when a star (yellow circle) is not obstructed by a planet (blue circle) and when a planet crosses the line of sight between the star and an observer. Suppose, both the scenarios are from an observer's (the Kepler Telescope, in our case) point of view.

At $t=0$, a planet is just about to cross in front, the star is not yet obstructed and we have a certain flux value F_1 . This value is a data point in Table I.

Now, at $t=1$, the planet comes into the line of sight, and so, from the perspective of the observer, we have a new (decreased) brightness value F_2 . This value $F_2 < F_1$ and is the value immediately to the right of F_1 in Table I. Fig. 3 represents the light curve for a single star over time.

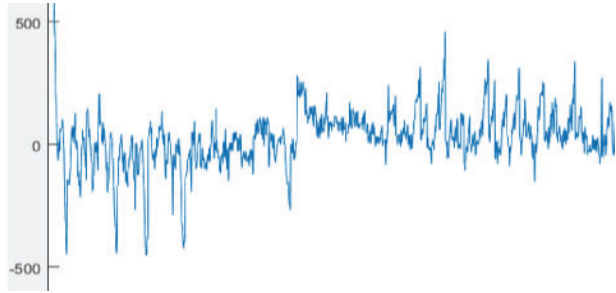


Fig. 3. Light curve for a single star plotted over time

The data mentioned here are based and originated from observations made by the NASA Kepler space telescope [18]. Our training set contains 5087 rows, where each row is an observation from an individual star and 3198 columns, where each column except is the flux value of that star recorded at that time. The first column is the label of the star and columns 2 - 3198 are the flux values over time. The training set is highly imbalanced as it has 37 positive and 5050 negative cases. Our test set contains 170 rows and 3198 columns, where the positive and negative classes are 15 and 155.

The reason behind this high imbalance of data is the fact that exoplanet systems are very rare. This gives us an insight into the difficulty of finding a positive case in large amounts of data and the need to evaluate new approaches to analyze them.

V. METHODOLOGY

A. Data Collection

The original data from NASA was collected, denoised and made publicly available on Kaggle[18], a platform for data science enthusiasts. The positive examples are labeled as 2 and the negative examples are labeled as 1.

99% of this data was obtained from campaign 3 of the ongoing Kepler mission. Since the number of confirmed exoplanets is very low, the dataset also contains confirmed cases from other campaigns. Specifically, the dataset includes all observations from Kepler's 3rd campaign, which makes up 99% of the data and the rest is obtained from other campaigns to ensure better class representation.

The dataset was obtained from the Mikulski Archive [19], an archive used by NASA to open source Kepler mission data.

After receiving the data from the telescope, NASA applies de-noising algorithms to isolate instrumental errors. The data is stored in. fits format in the archive and was obtained [18] with the help of a domain expert.

TABLE I. STRUCTURAL PRESENTATION OF THE RAW DATA

LABEL	FLUX1	FLUX2		FLUX3197
1	-0.729953	-0.415605		-0.0941791
1	-0.730652	-0.414605		-0.0875139
2	1.52615	0.215347		-0.0954629

B. Data Pre-Processing

To deal with the imbalanced data, we eliminated a large number of negative cases, so that the training data had a 62:38 class ratio and the testing data had a 59:41 class ratio.

Before applying feature extraction or classification algorithms on the data, it is important to take into consideration their relative scale and normalize them if necessary. We normalized the entire dataset before feature extraction process.

C. Feature Selection

Since ANFIS systems have restrictions on the number of inputs that they can take, we used various combinations of PCA, DTW and statistical measures but at most 6 of them were used as inputs at a time. The individual features are described in the following section and in the results section, we present three such combinations of features and compare the performance for each selection.

D. Feature Extraction

Although each data point of a time-series curve can be regarded as a feature, it is not suitable to use many features in a classification task. Attempting to do so causes various unwanted phenomena collectively labeled as 'the curse of dimensionality'.

To tackle this problem, the reduction of number of features while retaining the information that is needed to discriminate among samples is helpful. We define some features to substitute for the original data points of each time series curve.

1. Statistical Measures:

For each of our sample curves, we first obtained seven statistical features-mean, minimum, maximum, value of the 25th, 50th and 75th percentiles, and standard deviation. These features are calculated for each row, so, in essence, we calculate these features for the flux values of each star. Here, mean is the arithmetic mean of a time curve, or simply put is the sum of the values divided by the total number of flux values in that curve. Minimum and maximum are the lowest and highest peaks, respectively. 25th, 50th and 75th percentiles are the threshold values where, in order, 25%, 50% and 75% of the values are lower than it. Standard deviation is the amount by which the data points deviate from the mean, defined by the equation (1).

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{N-1}} \quad (1)$$

Now, since a transition event is likely to cause pronounced changes in brightness, we are interested in the highest and lowest peaks. Also, combining peak and percentile values with standard deviation will give us insight into whether a curve suddenly changes behavior for a short period and thus use these behaviors as features.

2. Principal Component Analysis:

Then, we use principal component analysis to reduce the number of features of each curve from 3198 to N. This value of N, known as the number of components is an important parameter of the PCA algorithm and so was chosen systematically. We solved the following equation to find an appropriate number of components.

Or phrased differently, we chose the highest value of k for which-

$$\frac{\frac{1}{m} \sum_{i=1}^m |x^{(i)} - x_{approx}^{(i)}|^2}{\frac{1}{m} \sum_{i=1}^m |x^{(i)}|^2} \leq 0.01 \quad (2)$$

This ensured that 99% of the variance of our data was preserved even after drastically reducing the number of features to $N=6$.

We also generated a 3 component representation of our data by setting the variance threshold at 90%.

3. Feature extraction by Dynamic Time Warp:

Let us consider curve 1 and curve 2 as rows in the data table
1. Comparing those curves using DTW will output a single number. For example in our data,

1. DTW (curve1, curve2) = 104.0204
2. DTW (curve5, curve7) = 239.0706
3. DTW (curve7, curve5) = 239.0706 [same as 2]
4. DTW (curve6, curve6) = 0 [distance between the same curves is zero]

Similarly, we have produced a distance matrix that has distances between all curves in the training data (Table II).

TABLE II. DTW DISTANCES FROM/TO ALL TRAINING EXAMPLES

	Curve1	Curve2	Curve3		Curve108
Curve1	0	104.02	176.32		104.99
Curve2	104.02	0	197.98		27.458
Curve3	176.32	197.98	0		191.28
.....					
Curve108	104.99	27.458	191.28		0

In general, DTW determines the optimal match between two sequences. It also has the ability to match between 2 sequences of different length, however, this feature is redundant in our application since we have curves of equal length.

As apparent from its definition, DTW is a comparison between two time series. We need to devise a way to use these similarity measures as features. To do so, we first isolate all the positive examples and calculate the DTW distances from all curves to *only* the ones with positive labels. We use these distances as features for each curve. The procedure can be expressed in the following pseudo code-

```
//F-DTW(i) is the set of features
For i := 1 to n //n=108, all examples
for j := 1 to m // m=38, +ve examples
    F-DTW(i) := F-DTW(i) .append(DTW(i, j))
```

So essentially, if T denotes each curve and $D = \{Q_1, Q_2, \dots, Q_M\}$ is the training data, extracted features, Feature-DTW(T), derived by DTW will be:

$$F\text{-DTW}(T) = \text{DTW}(T, Q_1), \text{DTW}(T, Q_2), \dots, \text{DTW}(T, Q_N)$$

where N is the number of positive examples and $N < M$.

And after the necessary calculations, Table 2 is reduced to Table III.

TABLE III. DTW DISTANCES FROM ALL TRAINING EXAMPLES TO ONLY POSITIVE EXAMPLES.

	Curve1	Curve2	Curve3		Curve38
Curve1	0	104.02	176.32		110.91
Curve2	104.02	0	197.98		38.19
Curve3	176.32	197.98	0		191.25
.....					
Curve108	104.99	27.458	191.28		29.125

It is notable that since all the positive cases will be compared to themselves, DTW will generate diagonal zeroes in the result matrix for only the positive cases. DTA will allow us to relate transition events in the form of similar 'shapes', even if they occur at different places on the time axis.

However, due to the strict limit on number of inputs to ANFIS, we take the arithmetic average of all the DTW distances of each curve to all the positive examples. Using pseudo code-

```
for i := 1 to n    //n =108, all examples
sum:=0
for j := 1 to m // m=38, +ve examples
    sum:=sum + DTW(i,j)
F-DTW(i) :=sum/m
```

So, Table III is further reduced to Table IV.

TABLE IV. AVERAGE DISTANCE TO POSITIVE CURVES

	Avg.DTW distance to positive curves
Curve1	676.669
Curve2	644.5492
.....	
Curve108	639.68

E. ANFIS classification

Our implementation of ANFIS was done through the neuro fuzzy software package available in Matlab. The system comprised of 6 inputs, a Sugeno type inference system and one output. Fig. 5 depicts our ANFIS structure.

The shape of the membership functions is chosen to be Gaussian and initialized to a default value. Then, we chose the

subtractive clustering method provided through the 'genfis2' command to create an initial fuzzy inference system. ANFIS considers training examples and modify the shape of the initially generated membership functions. The other alternative to subtractive clustering is grid partitioning. Grid partitioning suffers from the curse of dimensionality and so the system might run out of memory. Subtractive clustering overcomes this problem with no observed loss in performance. We chose the *area of influence* to be 0.25.

VI. RESULTS AND DISCUSSIONS

First we have provided our classifier with only statistical measures, namely: Mean, Standard Deviation, 25th. percentile, 50th. Percentile, 75th. percentile, Minimum and Maximum. The classifier did not perform satisfactorily based on these inputs. The training error has remained fairly constant throughout the 40 epochs that our classifier is trained. Both backpropagation and hybrid approach for optimization have produced similar results. The minimum RMS error after the final epoch is fairly large at .41 and the classification accuracy is close to 67%.

Then, we have used 6 principal components of the individual curves as features. It should be noted that 6 PCAs retained 99% of the variation of the data. As previously, we have used both hybrid and back-propagation optimization approaches to train our system for 40 epochs. The RMS error has stabled after about halfway through the training. We have achieved a slightly higher performance with principal components at a minimum RMS error of 34.3. The classification accuracy has also been improved to 74%.

Despite the increase in performance, the system still has performed below expectations with Principal Component Analysis. To overcome this problem, we chose a combination of statistical features (Mean, STD), principal components and the average of the Dynamic Time Warp Distances to each positive example as features. So, our 3rd. combination of features formed the feature vector-

[Mean Std. PC1 PC2 PC3 DTW]

The inclusion of DTW to our feature set has increased the performance significantly. The RMS error is dropped to around 0.29 we and a maximum accuracy of 81% is observed after multiple runs. Table V records the performance.

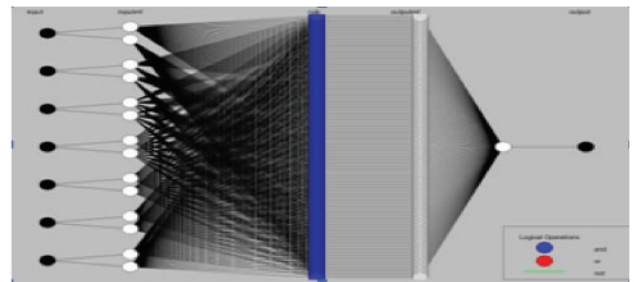


Fig. 4. ANFIS structure

TABLE V. COMPARISON OF FEATURE SETS IN CLASSIFICATION PERFORMANCE

Feature Vector	Minimum RMSE	Highest accuracy
[Mean StD 25% 50% min max]	0.41	~67%
[PC1 PC2 PC3PC4 PC5 PC6]	0.34	~74%
[Mean StD PC1 PC2 PC3 DTW]	0.29	~81%

The tuned Membership Functions for each of the 6 input variables after training is illustrated in Fig. 5:

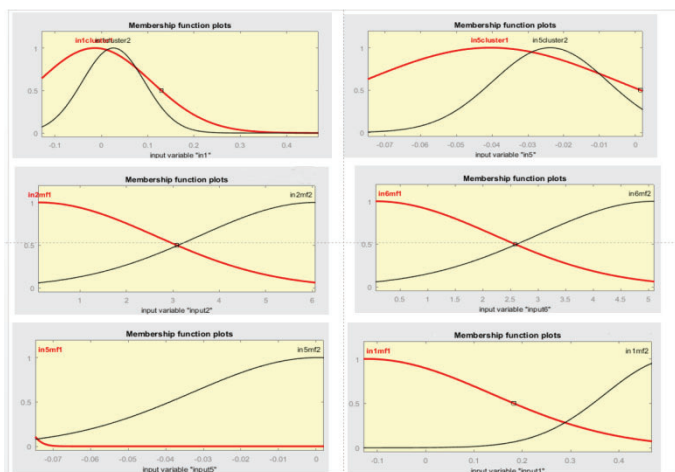


Fig. 5. Membership functions after training

Our Final ANFIS Structure relates all the internal nodes with the inputs and the output is given in Fig. 4.

We also summarize the performance of our classifier in the following Table VI.

TABLE VI. CONFUSION MATRIX ATTAINED FROM ANFIS PREDICTION.

	(Predicted) Class1	(Predicted) Class2
(Actual) Class1	74	3
(Actual) Class2	18	13

We use this data to calculate the following metrics (Table VII) for assessing the performance of our model-

TABLE VII. TABLE VII. ASSESSMENT METRICS

Accuracy	~81%
Misclassification Error	~19%
TPR	~42%
FPR	~23%
Prevalence	~29%

VII. CONCLUSIONS

In conclusion, the low dimensionality of the input makes ANFIS unsuitable for completely autonomous labeling of transition curves. Also, because of the under-representation of positive cases in such data where positive events are rare, it is hard to construct an accurate classifier using ANFIS.

That being said, the low false positive rate indicates that the Neuro-Fuzzy method can be competent to flag 'interesting' flux signatures for review by a human expert. In fact, almost three out of every four flagged time curve has proved to contain actual transition events.

VIII. FUTURE WORK

This work can be extended upon by evaluating the performances of other machine learning algorithms in classifying curves that contain transit events. Also the use of different feature extraction methods along with or independent of the ones used in this paper could be explored.

One classification approach that is particularly interesting is Deep Learning. Deep neural network architectures like Recurrent Neural Networks and Convolutional Neural Networks stand out as promising candidates. Recurrent Neural Networks, also known as RNNs, are of significance since this architecture takes into account the sequence of inputs. One possible approach can be to transform the input curves into sequences of smaller curves and then feed them into the classifier.

Also, access to a dataset with more positively labeled time curves will allow better classification results. The number of positive classes is very low since confirmed transition events are very rare.

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